

Last Mile

Manual Test Plan

Everything a manual tester needs: how to set up, what to check, how to check it, how to record results and report defects — for the collections decision engine, its Manager and Admin consoles, the daily cycle and the configuration editor.

System under test	Last Mile engine (port 8000) + simulated Riverbend Credit Union bank API (port 8001)	
Document version	1.0 · 15 September 2026	
Test cases	108 manual cases across 15 areas (55 priority P1)	
Estimated effort	2 days for one tester: day 1 areas ENV-REL, day 2 CLOSE-NFR and regression of defects	
Tester	_____	Build / date tested: _____
Tester signature & date	Developer lead	Product owner

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1 • What the system does

A bank already predicts which members may miss payments. **Last Mile** decides, every business day, **who the collections team should contact, how, and why** — within each collector's working hours and the bank's rules — and a **manager approves every action** before anything reaches a member.

1 • Team Manager enters who works today and for how long	2 • Run 8 AI agents read bank data, score accounts and build one queue per collector	3 • Review Manager approves, changes or rejects each recommendation	4 • Release Approved actions are sent to the bank	5 • Close day Report is frozen; the simulated bank moves to the next day
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PAGE	ADDRESS	WHO USES IT	WHAT IT IS FOR
Manager Console	127.0.0.1:8000/manager	Collections manager	Team, worklist, queues, decisions, release, close day
Daily report	127.0.0.1:8000/report	Manager, leadership	One page per business day
Admin Console	127.0.0.1:8000/admin	Admin, developer	Every stage, agent, tool call, table and LLM call
Configuration	127.0.0.1:8000/admin/config	Admin	Edit and validate the three YAML rule files
Guided tour	127.0.0.1:8000/guide	Clients	Plain-language walkthrough of a run
Bank API docs	127.0.0.1:8001/docs	Tester	See what the bank received and its business date

Words used in this document

Run	One execution of the agents. Has an id like <code>run_20260915T063217_3955</code> . Always note it in results.
Business day	The bank's date (<i>as of</i>). It only changes when the manager closes the day. It can be ahead of the real calendar.
Roster / team	Collectors, whether each is working, and shift minutes. 90% of each shift is plannable (10% buffer).
Queue	The ordered list of tasks for one collector. Texts (SMS) go to <i>Automated SMS</i> .
Escalated	A recommendation that needs individual review; excluded from bulk approve.

Superseded	A worklist replaced by a newer one for the same day (re-plan or new run). Read-only.
Segments	Persuadable (contact helps), Sure thing (pays anyway), Lost cause (unlikely to pay), Sleeping dog (contact may make it worse), Uncertain.

2 • Scope

In scope

- Team (roster) entry, validation and carry-forward
- Starting runs; the 8 stages; failures
- Worklist, per-collector queues, filters, account review
- Approve / change / reject / bulk approve
- Re-plan after a team change
- Release to the bank; locking
- Close the day; bank day advance; daily report
- Admin Console incl. LLM call records
- Configuration editor: check, save, history, conflicts
- Governance rules (suppression, contact caps, audit)
- Layout at phone width, dark mode, performance, error messages

Out of scope

- Accuracy of the statistical model (covered by automated tests and the benchmark)
- The /demo page — contains known outdated claims; not a release target
- Real bank integration, logins and permissions (none exist; everyone is *manager.demo* / *admin.demo*)
- Load testing, security penetration testing
- Wording quality of Gemini output beyond the checks in ACC-03/04

Priorities. **P1** a member could be wrongly contacted, data is wrong or lost, or the main flow is blocked. **P2** a feature does not work but there is a workaround. **P3** cosmetic or minor.

Automated tests exist. Developers run `make test` (39 tests). Manual testing focuses on what a person sees and does; if you find a rule broken that you think a test should catch, say so in the defect.

3 • Test environment

3.1 Prerequisites

Machine	macOS or Linux, 8 GB RAM, internet (only for Gemini)
Tools	<code>uv</code> (Python package manager), <code>make</code> , a terminal, Google Chrome (latest). Optional: <i>DB Browser for SQLite</i> for section 6.
Project	The <code>kamat</code> folder from the developer
Gemini key	Optional. Ask the developer; needed only for ADM-05/07 and LLM wording checks. Never paste it into a defect.

3.2 First-time setup

```

cd kamat
make setup          # installs Python 3.11 and dependencies, creates .env
# optional: open .env and set GEMINI_API_KEY=...
make seed           # creates the simulated bank (about 2 seconds)
make up             # starts bank (8001) and engine + consoles (8000); leave this terminal open

```

3.3 Daily start / stop

Start: `make up`. Stop: press **Ctrl+C** in that terminal. After the developer delivers a new build, **stop and start again** — the servers do not reload code on their own.

3.4 Reset to a clean state

Use before a full test cycle, or when data from earlier testing gets in the way.

```

Ctrl+C             # stop make up
make seed           # new simulated bank, business day = today's real date
make clean-runs     # deletes all runs, decisions, rosters, reports, audit and LLM records
make up

```

Reset deletes history. Export or screenshot anything you need as evidence first. `make clean-runs` does not reset configuration files — use the editor's *Earlier versions* to restore those (CFG-09).

3.5 Test data

All data is **synthetic**: 4,000 members, about 1,500 accounts behind on payments, 120 days of history. Names like *Morgan Nguyen* are fake. Generation is deterministic for a given date: two testers who reset on the same day see the same numbers. Figures quoted in this plan (e.g. 291 recommendations) are examples; yours will differ.

4 • How to run, record and track testing

4.1 Order of execution

Execute areas in this order — later areas need state created by earlier ones: **ENV** → **TEAM** → **RUN** → **WL** → **ACC** → **DEC** → **RPL** → **REL** → **CLOSE** → **RPT** → **ADM** → **CFG** → **GOV** → **OTH** → **NFR**. Run the full **P1** set first on day 1; P2/P3 on day 2.

4.2 Recording a result

1. Tick **Pass** only if every bullet under Expected is true.
2. Tick **Fail** if any bullet is false; raise a defect (template 8.1) and write its id in the Defect line.
3. Tick **Blocked** if you could not execute (e.g. bank would not start); note the blocking defect id.
4. Leave untested cases blank; they count as *Not run*.
5. For every run you create, write the **run id** next to the case (shown in the Manager chips and Admin banner).

4.3 Tracking sheet

Keep one spreadsheet with one row per case. Columns:

COLUMN	CONTENT
Case ID	e.g. DEC-02
Priority	P1 / P2 / P3

COLUMN	CONTENT
Status	Pass · Fail · Blocked · Not run · Retest pass · Retest fail
Run id / business date	run_... and the bank date shown in the Manager panel
Defect ID	e.g. LM-014
Evidence	Screenshot file name(s)
Tested on / by	Date, tester initials, build
Notes	Anything unexpected even if passed

4.4 Progress summary

AREA	NAME	CASES	P1	PASS	FAIL	BLOCKED	NOT RUN
ENV	Setup and health	7	3				
TEAM	Today's team (roster)	10	3				
RUN	Starting a run	7	5				
WL	Worklist and collector queues	8	3				
ACC	Account review	8	5				
DEC	Decisions	8	5				
RPL	Re-plan when the team changes	6	4				
REL	Release to the bank	4	3				
CLOSE	Close the day	6	4				
RPT	Daily report	8	5				
ADM	Admin Console	8	3				
CFG	Configuration editor	13	7				
GOV	Governance rules	6	5				
OTH	Guided tour, comparison and model pages	4	0				
NFR	Usability, layout and performance	5	0				
	Total	108	55				

4.5 Collecting evidence

Screenshot	Whole browser window including the address bar. Name: <code>CASEID_short-description.png</code> .
Run id	Manager top chips or Admin run banner.
Browser errors	Chrome → View → Developer → JavaScript Console. Copy red lines.
Server errors	The <code>make up</code> terminal. Copy the traceback (lines starting <i>Traceback</i> down to the error).
What an agent did	Admin → stage → <i>Agent & tool calls</i> ; click a row for input/output.
Who did what, when	Manager → Audit trail tab.

What the bank received	127.0.0.1:8001/docs → GET /api/v1/collections/actions.
LLM request/response	Admin → LLM calls panel.

5 • Test cases

Each case: ID + priority · what to check and steps · expected result (all bullets must hold) · result.

ENV Setup and health

Confirms the environment is correct before any functional testing. If an ENV case fails, stop and report it: every later result depends on it.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
ENV-01 P1	Both services start 1. In a terminal in the kamat folder run <code>make up</code> . 2. Wait until the URL list is printed.	- Terminal lists Bank API :8001, Guided tour, Demo, Admin, Manager, Daily report and Configuration URLs. - No red traceback in the terminal.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ENV-02 P1	Bank API is reachable 1. Open <code>http://127.0.0.1:8001/docs</code> . 2. Expand GET /api/v1/health → Try it out → Execute.	Response 200 with <code>status: ok</code>, institution <i>Riverbend Credit Union</i> and an <code>as_of</code> date.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ENV-03 P1	Sign in and read the app shell 1. Open <code>http://127.0.0.1:8000/manager</code> and sign in with the account you were given.	- Left sidebar: Today, Daily reports, Runs & agents, Configuration, Guided tour, Bank API; the current page is highlighted. - Top bar: page title, chip <i>Bank</i> · <code><date></code> with a green dot, the LLM chip, and a <i>cfg</i> hash chip. - Top right: your username and role, and a menu with Light / Dark / System and Sign out.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ENV-06 P2	Roles show only what they may do 1. Sign in as <i>guest</i> , open <code>/manager</code> , then <code>/admin/config</code> . 2. Sign out, sign in as <i>manager</i> , open <code>/admin/config</code> .	- Guest: no Approve, Release, Start run, Close day or Save buttons; a note says the account is read-only. - Manager: the configuration editor loads and Check changes works, but Save... is not shown. - Admin: everything is available.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ENV-07 P3	Theme and sidebar preferences stick 1. Switch the theme to Dark in the user menu; collapse the sidebar; reload; open another console.	Both choices persist across pages and reloads.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ENV-04 P2	Bank down is reported clearly 1. Stop only the bank (or stop <code>make up</code> and start <code>make engine</code> alone). 2. Open <code>/manager</code> and <code>/admin</code> .	- Manager shows a red callout <i>The bank is not reachable</i> with the reason. - The top-bar chip reads <i>Bank unreachable</i> with a red dot on every console. No blank page, no crash. - Restart <code>make up</code> afterwards.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
ENV-05 P3	All pages load Open /guide, /demo, /admin, /manager, /report, /admin/config.	- Each page loads without a browser error. Open DevTools → Console: no red errors. - The four operator consoles share the same sidebar and top bar; /guide and /demo keep their presentation layout.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

TEAM Today's team (roster)

The manager tells the planner who is working today and for how long. Location: Manager Console → top panel → **Confirm today's team / Edit today's team.**

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
TEAM-01 P1	Default team on a fresh day - Reset to a clean state (section 3.4). - Open /manager.	- Step 1 <i>Confirm today's team</i> is highlighted. - Callout says no team has been saved and the default team is shown. - Panel shows 5 collectors · 1,800 min.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
TEAM-02 P1	Edit and save a team - Click Confirm today's team. - Rename Collector 1 to <i>Priya</i>; set Collector 2 shift to <i>Part time · 4 h</i>; switch Collector 3 off. - Type a note; click Save today's team.	- Toast <i>Today's team saved</i>. - Tiles show Working 4 of 5, Shift minutes 1,320, Plannable 1,188. <i>Saved today</i> table lists the save with time, 4 working, 1,320 minutes and the note. - Step 1 turns green (✓).	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
TEAM-03 P2	Plannable minutes = 90% of shift In the team drawer, set a collector to Custom 200 minutes.	Plannable for that row shows 180 min. Tile totals update immediately.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
TEAM-04 P2	Custom minutes are bounded - Enter 900 in Minutes; tab out. - Enter -30; tab out.	Value is clamped to 720 and 0 respectively.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
TEAM-05 P2	Name is required Clear a collector's name; click Save.	Red toast <i>Every collector needs a name</i>. Nothing is saved.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
TEAM-06 P2	At least one collector Remove every row with X; click Save.	Red toast <i>Add at least one collector</i>.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
TEAM-07 P2	Nobody working Switch every collector off; Save.	- Amber warning <i>Nobody is working: only automated text reminders can be planned</i>. Save is allowed. - A run started afterwards produces only SMS tasks (check in TEAM queues).	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
TEAM-08 P2	Add collector gets a unique id 1. Click + Add collector twice; save; reopen.	New rows named Collector 6, Collector 7. Saving succeeds (no duplicate-id error).	☐ Pass ☐ Fail ☐ Blocked Defect
TEAM-09 P1	Team carries forward to the next day 1. Save a non-default team, run the day through Close the day (CLOSE-01). 2. On the new business day open the team drawer.	- Callout says the team is carried forward from the previous date. - Same collectors and shifts. Step 1 is <i>todo</i> until saved again.	☐ Pass ☐ Fail ☐ Blocked Defect
TEAM-10 P3	Start with this team 1. On a day with no saved team click Start with this team .	The shown team is saved automatically (note <i>confirmed at run start</i>) and the run starts.	☐ Pass ☐ Fail ☐ Blocked Defect

RUN Starting a run

The agents pull the bank's data, score every account and build the collector queues. A run takes about 10 seconds without an LLM, 50–70 seconds with Gemini.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
RUN-01 P1	Start today's run from Manager 1. After saving the team click Start today's run .	- Toast <i>Started run_...</i> - Button changes to <i>Agents are planning...</i> with a <i>Watch in Admin Console</i> link. - When finished: toast <i>Today's worklist is ready</i>, worklist loads, step 2 ✓.	☐ Pass ☐ Fail ☐ Blocked Defect
RUN-02 P1	Stages complete in Admin 1. Click <i>Watch in Admin Console</i> during a run.	- Eight stage cards fill in order: Data ingestion → Configuration → Data quality & privacy → Feature layer → Decision engine → Policy & governance → Explanation → Human approval. - First seven show green; Human approval shows amber (waiting). - Live trace on the right keeps scrolling; run banner shows status <i>awaiting approval</i>.	☐ Pass ☐ Fail ☐ Blocked Defect
RUN-03 P1	Run uses the saved team 1. Admin → select the run → Configuration stage → Outputs.	- Table <i>Today's team (Manager Console roster)</i> lists exactly the collectors, working flags and shifts you saved. <i>Plannable minutes</i> metric = sum of working shifts × 0.9.	☐ Pass ☐ Fail ☐ Blocked Defect
RUN-04 P1	No double start 1. Click Start, then immediately refresh /manager and look for the Start button.	- While running, only <i>Agents are planning...</i> is shown (no second Start). - Calling Start again via Admin → Start new run gives red toast <i>run ... is still running</i>.	☐ Pass ☐ Fail ☐ Blocked Defect

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
RUN-05 P2	Second run the same day replaces the first 1. With a worklist awaiting approval (nothing released), Admin → Start new run.	New run reaches awaiting approval. The earlier run shows status superseded in Admin's run list and disappears from Manager's run list.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
RUN-06 P1	No run after release on the same day 1. Release at least one action (REL-01). Admin → Start new run.	Red toast: <i>today's actions (<date>) were already released ... close the day in the Manager Console.</i>	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
RUN-07 P2	Failed run is visible 1. Stop only the bank (run <code>make engine</code> alone); Admin → Start new run.	Run ends failed at Data ingestion with a red error in the run banner. No worklist is published; Manager does not list the run.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

WL Worklist and collector queues

Location: Manager Console → Worklist and Team queues tabs.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
WL-01 P1	Summary tiles 1. After a run, read the tiles above the tabs.	Recommendations, Est. avoided loss (with P10–P90 range), Team today (N collectors, minutes used of plannable), Escalated, Decisions 0 / N, Released 0.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
WL-02 P1	Every task belongs to a working collector 1. Worklist → Collector filter → pick each collector in turn.	- Voice tasks (Collector call, Payment plan offer, Hardship referral) are only assigned to collectors who are working. - SMS tasks are under <i>Automated SMS</i>. - No collector who was switched off appears in the filter.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
WL-03 P1	Queue fits the shift 1. Team queues tab: read each collector card.	$planned\ min \leq plannable\ min$ for every collector (e.g. 322 of 324). - No card shows <i>over shift</i> before any decisions are made.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
WL-04 P2	Queue order 1. Worklist → filter one collector.	Rows are ordered #1, #2, #3... and Value decreases down the list.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
WL-05 P2	Review this queue 1. Team queues → Review this queue → on one card.	Switches to Worklist filtered to that collector; count shows e.g. <i>16 of 291</i>.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
WL-06 P2	Filters combine 1. Set Action = CALL and Segment = Persuadable and tick <i>escalated only</i> .	Only rows matching all filters; count updates. Clearing filters restores all rows.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
WL-07 P2	Escalation flags 1. Filter <i>escalated only</i> .	Each row shows at least one flag: HIGH EXPOSURE (balance $\geq$ \$30,000), WIDE INTERVAL, or HARDSHIP OFFER (every hardship referral).	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
WL-08 P3	Simulated completion shown 1. Team queues $\rightarrow$ each card.	<i>Simulated day: NN% of this queue finishes in the shift</i> is present for working collectors (typically 90–100%).	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

ACC Account review

Click any worklist row. A drawer opens on the right.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
ACC-01 P1	Recommendation card 1. Open a row.	Shows action, rank, segment pill, template source (LLM template / fallback template), flags, and tiles Action value, Uplift (with range), Expected loss, Base cure.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ACC-02 P1	Every figure in the explanation is sourced 1. Tick show sources . Hover figures in <i>Why</i> .	- Every number is highlighted and shows a source such as <code>bank:accounts.CURR_BAL</code>, <code>model:tlearner-...</code>, <code>solver:cbc</code>. Fail if any number in the sentence is plain (not highlighted).	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ACC-03 P1	Explanation reads correctly 1. Read the <i>Why</i> paragraph for 5 different segments.	Grammatical English; no <i>a auto loan</i>, no <i>an \$9,894 of \$9,894</i>; the sentence fits the segment (e.g. a Sleeping dog says contact may backfire).	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ACC-04 P1	Outreach script 1. Read <i>Outreach script</i> .	Uses the member's first name; warm tone; no internal words (uplift, model, risk grade, <i>Collector call</i>).	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ACC-05 P2	Policy checks 1. Scroll to <i>Policy checks</i> .	Every suppression flag, the 7-day contact cap and eligibility rules listed with $\checkmark$ / X. The recommended action's rules all pass.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
ACC-06 P2	Why this account (counterfactual) 1. Click Ask the Decision Agent .	- Green callout: dropping this account lowers the plan by \$X, followed by the accounts that would replace it. - Admin trace shows a HANDOFF from Collections manager and an <code>engine.explain_selection</code> call.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
ACC-07 P2	Not-selected account 1. Compare tab → a row under <i>Chosen by the risk sort, skipped by the optimiser</i> → Ask the Decision Agent.	Explains why it lost: outcompeted (with cost), no positive value, or no allowed action.	☐ Pass ☐ Fail ☐ Blocked Defect
ACC-08 P1	No personal data shown except first name 1. Read <i>Account data (tokenised)</i> .	Account and member shown only as tokens (acc_..., mem_...). No surname, phone or email anywhere in the drawer.	☐ Pass ☐ Fail ☐ Blocked Defect

DEC Decisions

Location: account drawer → *Your decision*, and the buttons above the tabs.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
DEC-01 P1	Approve one 1. Open a pending row → Approve <action> .	Toast; row decision pill = APPROVED; Decisions tile increments; drawer shows <i>Current: approved by manager.demo</i>.	☐ Pass ☐ Fail ☐ Blocked Defect
DEC-02 P1	Change action needs a reason 1. Pick a different action in the dropdown, leave Reason empty → Change action .	Red toast <i>a reason code is required to edit or reject a recommendation</i>.	☐ Pass ☐ Fail ☐ Blocked Defect
DEC-03 P1	Change action with reason 1. Pick action + reason (e.g. WRONG_ACTION) → Change action.	Pill EDITED with → NEWACTION. Audit trail tab shows <code>approval.edited</code> with the reason.	☐ Pass ☐ Fail ☐ Blocked Defect
DEC-04 P1	Reject needs a reason 1. Reject with empty reason; then with reason TONE.	First: red toast. Second: pill REJECTED; audit entry <code>approval.rejected</code>.	☐ Pass ☐ Fail ☐ Blocked Defect
DEC-05 P2	Only allowed actions offered 1. For an account without phone consent (Policy checks show X phone_consent), open Change action dropdown.	Voice actions are not offered. Only allowed actions appear.	☐ Pass ☐ Fail ☐ Blocked Defect
DEC-06 P1	Bulk approve skips escalated 1. Click Approve all non-escalated (N) .	Toast <i>Approved N; M escalated left for review</i>. Escalated rows stay PENDING.	☐ Pass ☐ Fail ☐ Blocked Defect
DEC-07 P2	Edits that overfill a shift are flagged 1. Change several CALL (12 min) tasks of one collector to Hardship referral (45 min) where allowed.	That collector's card in Team queues shows over shift and an amber outline once committed + pending minutes exceed plannable.	☐ Pass ☐ Fail ☐ Blocked Defect

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
DEC-08 P2	Decisions persist 1. Refresh the page.	All decisions remain; counts identical.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

RPL Re-plan when the team changes

If staffing changes after the worklist was produced, the manager re-plans. Data and model are reused, so it takes seconds.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
RPL-01 P1	Banner after team change 1. With a worklist awaiting approval, Edit today's team → switch someone on/off or change a shift → Save.	- Drawer shows amber <i>Today's worklist was planned for a different team</i> with Re-plan now. - Top panel shows amber banner and primary Re-plan for the new team.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
RPL-02 P1	Re-plan produces a new worklist 1. Click Re-plan (confirm if asked).	<i>Agents are planning...</i>, then a new run id loads. - Team queues now match the new team; absent collectors have 0 tasks. - Admin: stages Data ingestion, Data quality and Feature layer say <i>Reused from run ...</i>; model id is the same as the old run.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
RPL-03 P1	Old worklist is locked 1. In Admin select the old run; note its status. Try approving in it (Manager ?run=<old id>).	Status superseded. Approve gives red toast <i>this worklist was replaced by ... after the team changed</i>.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
RPL-04 P2	Decisions warning 1. Approve 3 rows, change team, click Re-plan.	Confirm dialog says the 3 decisions will need to be made again.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
RPL-05 P2	Saving the same team does not ask to re-plan 1. With a worklist awaiting approval, open the team drawer and save without changing anything.	No re-plan banner in the drawer or the top panel. Existing decisions are untouched.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
RPL-06 P1	No re-plan after release 1. Release some actions; change the team.	Re-plan is refused: <i>some actions were already sent to the bank; re-planning could contact members twice</i>.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

REL Release to the bank

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
REL-01 P1	Release approved 1. Approve some rows → Release approved to bank (N) → OK.	- Toast <i>Released N actions to the bank</i>; rows show SENT; Released tile = N; step 4 ✓. - Bank docs → GET /api/v1/collections/actions lists N items with ACTION_CD, SCRIPT_TXT, APPROVED_BY.	☐ Pass ☐ Fail ☐ Blocked Defect
REL-02 P1	Released decisions are locked 1. Open a SENT row.	<i>Released to the bank — decisions are locked</i>; no approve/change/reject controls.	☐ Pass ☐ Fail ☐ Blocked Defect
REL-03 P2	Nothing to release 1. With no approved unreleased rows, the Release button.	Disabled, count 0.	☐ Pass ☐ Fail ☐ Blocked Defect
REL-04 P1	Script sent uses first name only 1. Bank docs → collections/actions → read SCRIPT_TXT.	Contains first name and institution name; no surname, phone or model terms.	☐ Pass ☐ Fail ☐ Blocked Defect

CLOSE Close the day

Freezes the day's report and moves the simulated bank to the next business day. Started only by the manager.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
CLOSE-01 P1	Close a completed day 1. After releasing, click Close the day → confirm.	- Toast <i>Closed <date>. The bank is now on <date+1>.</i> - Offer to open the report. - Top panel shows the next date with step 1 todo and team carried forward.	☐ Pass ☐ Fail ☐ Blocked Defect
CLOSE-02 P1	Warn about unreleased or undecided 1. With pending (e.g. escalated) or approved-but-unreleased rows, click Close the day.	Second dialog: <i>N approved action(s) have not been released and M recommendation(s) were never decided; they will not happen. Close anyway?.</i> Cancel keeps the day open.	☐ Pass ☐ Fail ☐ Blocked Defect
CLOSE-04 P1	Bank really moved on 1. Bank docs → GET /api/v1/health; GET /api/v1/simulation/day-log.	as_of is the next date. Day log row for the closed date: ENGINE_ACTIONS and EXECUTED equal the released count.	☐ Pass ☐ Fail ☐ Blocked Defect
CLOSE-05 P1	Closed day is locked 1. Open Manager with ?run=<closed day's run id> → Release.	Red toast <i>business day ... is closed; its actions can no longer be released.</i>	☐ Pass ☐ Fail ☐ Blocked Defect

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
CLOSE-06 P2	Bank error does not half-close 1. Stop the bank; click Close the day.	Red toast <i>the bank did not close the day....</i> Day stays open; retrying after restart works.	☐ Pass ☐ Fail ☐ Blocked Defect
CLOSE-07 P2	Next day's data changed 1. Start a run on the new day; compare Admin → Data ingestion rows with the previous day.	Queue history row count grew; outcomes grew; accounts behind today differs slightly.	☐ Pass ☐ Fail ☐ Blocked Defect

RPT Daily report

Location: [/report](#) or Manager → **Daily report** / **Today's report**.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
RPT-01 P1	Live report while the day is open 1. Open today's report before closing.	- Amber pill Live: <i>the day is still open — this report updates as decisions are made.</i> - Approve one more row in Manager, refresh report: Decided count increases.	☐ Pass ☐ Fail ☐ Blocked Defect
RPT-02 P1	Final report after close 1. Open the closed date.	Green Final pill with closed by and time. Numbers no longer change when you refresh.	☐ Pass ☐ Fail ☐ Blocked Defect
RPT-03 P1	Totals match Manager 1. Compare report tiles with Manager tiles for the same run.	Recommended, approved, changed, rejected, undecided, sent, estimated avoided loss are identical.	☐ Pass ☐ Fail ☐ Blocked Defect
RPT-04 P1	Team and queues table 1. Read <i>Team and queues</i> .	One row per collector (including those not working) plus Automated SMS; shifts, planned and plannable minutes and decision counts per person match the Team queues tab.	☐ Pass ☐ Fail ☐ Blocked Defect
RPT-05 P2	Runs this day 1. After a re-plan day, read <i>Runs this day</i> .	Lists every run; superseded ones say <i>replaced by ...</i>; the re-plan says <i>re-plan of</i>	☐ Pass ☐ Fail ☐ Blocked Defect
RPT-06 P1	What happened next 1. Open yesterday's (closed) report after the day is closed.	<i>Carried out N / N; Reached the member percentage; Caught up within 30 days: not yet known</i> with the date outcomes arrive (business date + 30).	☐ Pass ☐ Fail ☐ Blocked Defect
RPT-07 P2	Day by day table 1. Scroll to the bottom; click another day.	All days listed with status open/closed; clicking loads that day and updates the URL.	☐ Pass ☐ Fail ☐ Blocked Defect

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
RPT-08 P3	Print 1. Click Print / save as PDF .	Print preview hides the top bar; sections are not cut mid-card.	☐ Pass ☐ Fail ☐ Blocked Defect

ADM Admin Console

Location: `/admin`. For checking what each agent did.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
ADM-01 P1	Stage detail tabs 1. Click each stage card; open Outputs, Agent & tool calls, Data.	Outputs: metrics and tables. Calls: every event with kind, agent, tool, module, ms. Data: tables with row counts.	☐ Pass ☐ Fail ☐ Blocked Defect
ADM-02 P1	PII is masked 1. Data ingestion → Data → <code>landing_members</code> .	first/last name, phone, email columns show <code>●●● masked (PII)</code>.	☐ Pass ☐ Fail ☐ Blocked Defect
ADM-03 P2	Trace filter 1. Click an agent chip (e.g. Policy) above the live trace.	Only that agent's events and handoffs to it are shown.	☐ Pass ☐ Fail ☐ Blocked Defect
ADM-04 P2	Ownership enforced 1. Agents panel at the bottom.	Each of 8 agents lists its tools; Tool registry shows owner per tool. No tool owned by the Explanation Agent is an engine.* tool.	☐ Pass ☐ Fail ☐ Blocked Defect
ADM-05 P1	LLM calls panel (Gemini key set) 1. Select a run made with Gemini → LLM calls panel → click a row.	1–2 calls (draft; rewrite if drafts failed checks). Detail shows status, model, HTTP 200, latency, tokens, system instruction, prompt, raw response, parsed JSON. - Prompt contains no account token (acc...), name, balance or number from an account.	☐ Pass ☐ Fail ☐ Blocked Defect
ADM-06 P2	LLM calls panel (no key) 1. Run with no GEMINI_API_KEY.	Panel says <i>No LLM calls in this run — it used template (...)</i>. Run still completes; templates marked fallback.	☐ Pass ☐ Fail ☐ Blocked Defect
ADM-07 P2	LLM failure falls back 1. Put an invalid GEMINI_API_KEY in .env, restart, run.	Run completes. LLM calls panel shows status failed with HTTP 400/403 and error text. Worklist uses fallback templates.	☐ Pass ☐ Fail ☐ Blocked Defect

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
ADM-08 P3	Links 1. Click Configuration, Manager Console, Demo.	Each opens the right page.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

CFG Configuration editor

Location: Admin → **Configuration** (/admin/config). **Always restore any change you save during testing** (CFG-09).

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
CFG-01 P1	Files and guide 1. Open the page.	- Left: Scenario Pack (collections_delinquency <i>in use</i>, npa_early_warning <i>not used</i>), Institution Pack, Run Config. - Section chips with coloured risk dots; guide panel explains the section.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
CFG-02 P1	Guide follows the cursor 1. Click inside <code>capacity:</code> on the line <code>planning_buffer</code> .	Guide shows <i>Team size and daily limits</i>, Read by chips, and highlights <code>planning_buffer</code> with its meaning.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
CFG-03 P1	Schema error with line 1. Change <code>planning_buffer: 0.10</code> to <code>0.7</code> → Check changes .	Red 1 <i>problem(s) must be fixed before saving</i>; message <i>Input should be less than 0.5</i>; a <i>line NN</i> link jumps to the line; gutter number red; Save disabled.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
CFG-04 P1	YAML syntax error 1. Remove one space of indentation under <code>capacity:</code> → Check.	<i>YAML syntax error</i> with a line number and advice about indentation.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
CFG-05 P1	Error across files 1. Rename the rule set <code>sms_consent:</code> to <code>sms_ok:</code> → Check.	<i>Across packs error</i>: action SMS requires unknown eligibility rules ['sms_consent'].	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
CFG-06 P2	Unknown data field is warned 1. Change a rule's <code>field: dpd</code> to <code>field: days_late</code> → Check.	Valid but a warning: <i>rule field 'days_late' is not a known data field; the next run will stop</i>.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>
CFG-07 P2	Engine compatibility 1. Scenario Pack: change <code>id: CALL</code> to <code>id: VISIT</code> → Check.	Error <i>Engine compatibility</i>: action id(s) ['VISIT'] are not known... Save disabled.	☐ Pass ☐ Fail ☐ Blocked <u>Defect</u>

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
CFG-08 P1	Valid change and save 1. Set planning_buffer 0.12 → Check → Save... → type a reason → Save.	- Check lists the change <i>0.1</i> → <i>0.12</i> with meaning and effect. Save needs a reason (fewer than 5 characters is refused). - Toast <i>Saved. The next run uses configuration ...</i>; hash chip changes; <i>1 earlier version(s)</i>. - Next run: Configuration stage shows plannable minutes at 88%.	☐ Pass ☐ Fail ☐ Blocked Defect
CFG-09 P1	Restore a version 1. Earlier versions... → pick the version → Check → Save with reason <i>restore after test</i> .	Editor loads the older text; check passes; after saving the file is back to the original value.	☐ Pass ☐ Fail ☐ Blocked Defect
CFG-10 P2	Edited in two tabs 1. Open the same file in two tabs; save a change in tab A; save a different change in tab B.	Tab B is refused: <i>this file changed on disk after you opened it; reload it and apply your edit again</i>.	☐ Pass ☐ Fail ☐ Blocked Defect
CFG-11 P2	No save during a run 1. Start a run; immediately try to save a valid change.	Refused: <i>run ... is in progress; it would stop on a configuration change</i>.	☐ Pass ☐ Fail ☐ Blocked Defect
CFG-12 P3	Unsaved edits guard 1. Edit text, then click another file or close the tab.	Confirmation asking to discard edits. Discard edits restores the file text.	☐ Pass ☐ Fail ☐ Blocked Defect
CFG-13 P2	Config change is audited 1. After CFG-08, run database check G4 (section 6).	Latest events include <i>config.changed</i> by the name you entered. Manager → Audit trail still shows <i>Hash chain verified</i>.	☐ Pass ☐ Fail ☐ Blocked Defect

G0V Governance rules

These protect members. A failure here is always P1.

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
G0V-01 P1	Suppressed members never recommended 1. SQL check G1 in section 6 (or ask a developer).	Zero recommended accounts with deceased, bankruptcy, cease-and-desist, litigation, fraud hold or do-not-call.	☐ Pass ☐ Fail ☐ Blocked Defect
G0V-02 P1	One contact per member per day 1. SQL check G2.	No member appears with more than one contact action in a run's worklist.	☐ Pass ☐ Fail ☐ Blocked Defect

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
G0V-03 P1	Zero violations 1. Admin → Policy & governance stage.	Metric <i>Violations (optimised)</i> 0; trace note <i>Optimised plan independently verified: zero violations</i>.	☐ Pass ☐ Fail ☐ Blocked Defect
G0V-04 P1	No number written by the AI 1. Manager → Model health → Governance tiles.	<i>Unstamped numbers</i> 0; <i>Provenance complete</i> N / N.	☐ Pass ☐ Fail ☐ Blocked Defect
G0V-05 P1	Audit chain intact 1. Manager → Audit trail.	Green <i>Hash chain verified</i>; events for published, approvals, releases, day closed, roster saved.	☐ Pass ☐ Fail ☐ Blocked Defect
G0V-06 P2	Hardship always escalated 1. Filter Action = HARDSHIP.	Every row carries the HARDSHIP OFFER flag and is skipped by bulk approve.	☐ Pass ☐ Fail ☐ Blocked Defect

OTH Guided tour, comparison and model pages

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
OTH-01 P2	Guided tour replay 1. Open /guide → Replay the latest run .	Stages light up in order with plain-language feed lines; chapters 4–7 fill with numbers from the run.	☐ Pass ☐ Fail ☐ Blocked Defect
OTH-02 P3	Technical detail toggle 1. Toggle Technical detail .	Tool names and module names appear/disappear; setting survives a refresh.	☐ Pass ☐ Fail ☐ Blocked Defect
OTH-03 P2	Sort vs Optimised and benchmark 1. Manager → Sort vs Optimised → Run synthetic benchmark.	Three policy cards; after benchmark: true value tiles, lift vs risk sort, capture of oracle, estimate ÷ truth ratio.	☐ Pass ☐ Fail ☐ Blocked Defect
OTH-04 P3	Model health 1. Manager → Model health.	Qini per action, decile table with action selector, feasibility checks, solver details.	☐ Pass ☐ Fail ☐ Blocked Defect

ID	CHECK AND STEPS	EXPECTED RESULT	RESULT
NFR-01 P2	Phone width 1. DevTools device toolbar 375 px wide: /manager, /report, /admin/config, /guide.	No sideways page scroll (tables may scroll inside their box); buttons usable; team drawer rows stack.	☐ Pass ☐ Fail ☐ Blocked Defect
NFR-02 P3	Dark mode 1. Switch OS to dark mode; reload pages.	All text readable; no white-on-white or black-on-black.	☐ Pass ☐ Fail ☐ Blocked Defect
NFR-03 P2	Run duration 1. Time RUN-01 with and without Gemini.	Without LLM ≤ 30 s; with Gemini ≤ 120 s. Re-plan ≤ 15 s without LLM.	☐ Pass ☐ Fail ☐ Blocked Defect
NFR-04 P3	Keyboard 1. Tab through the team drawer and config editor; press Esc on drawers.	Visible focus ring; Esc closes drawers; Tab inside the YAML editor inserts two spaces; Ctrl/Cmd+Enter runs Check.	☐ Pass ☐ Fail ☐ Blocked Defect
NFR-05 P2	Errors are understandable 1. Collect every red toast seen during testing.	Each says what went wrong and what to do. Log any toast that is raw JSON, a stack trace or unclear.	☐ Pass ☐ Fail ☐ Blocked Defect

6 • Optional database checks

For testers comfortable with SQL. Open the files read-only in *DB Browser for SQLite* (File → Open Database Read Only). **Never edit these files** — the audit tables refuse changes by design and editing others corrupts test data. Replace **RUN** with a run id.

Engine database	kamat/data/engine/lastmile.db
Bank database	kamat/data/bank/bank.db

G1 Suppressed accounts in a worklist (expect 0)

```
-- engine DB: data/engine/lastmile.db. Replace RUN.  
SELECT COUNT(*) FROM recommendations  
WHERE run_id = 'RUN' AND alternatives_json LIKE '%suppressed:%';
```

G2 Members with more than one contact (expect no rows)

```
SELECT member_token, COUNT(*) n FROM recommendations  
WHERE run_id = 'RUN' GROUP BY member_token HAVING n > 1;
```

G3 Minutes per collector vs shift

```
SELECT collector_id, COUNT(*) tasks,  
       SUM(json_extract(facts_json, '$.minutes.value')) minutes  
FROM recommendations WHERE run_id = 'RUN' GROUP BY collector_id;
```

G4 Latest audit events

```
SELECT seq, run_id, event_type, actor, created_at FROM audit_events ORDER BY seq DESC LIMIT 20;
```

G5 LLM calls of a run

```
SELECT id, purpose, status, http_status, latency_ms, total_tokens FROM llm_calls WHERE run_id =  
'RUN';
```

G6 Bank: actions and how they went (bank DB: data/bank/bank.db)

```
SELECT ACTION_CD, EXECUTION_STATUS, COUNT(*), SUM(RPC_IND)  
FROM collection_actions GROUP BY 1, 2;
```

7 • Known behaviour that is not a bug

YOU MAY SEE	WHY
Business date ahead of the real calendar	Every <i>Close the day</i> moves the simulated bank forward one day, regardless of the clock.
Most tasks are SMS; few hardship referrals	SMS uses no collector time. Payment plans are usually worth more per minute than hardship referrals.
Planned minutes a few below plannable (e.g. 322 of 324)	Tasks are 12, 25 or 45 minutes; the leftover minutes cannot fit another task.
Run takes about a minute with Gemini	Each Gemini call takes 20–25 seconds; there can be two (draft + rewrite).
Some explanations say <i>fallback template</i>	Gemini output that fails the checks, or no key, uses standard wording by design.
<i>Caught up within 30 days: not yet known</i>	Outcomes only exist 30 business days after the actions.
Estimated dollar values look large versus the benchmark	Model estimates are about 2× optimistic; this is documented. Do not log it as a defect.
Re-planned run reuses stages 1, 3 and 4 with 0 ms	Data and model are copied from the parent run on purpose.
Everyone appears as manager.demo / admin.demo	There is no login in this build.
/demo page claims (e.g. fixed ROI, “0 harmful”)	Known outdated page, out of scope.

Always a P1 defect, even if it looks small: a suppressed (deceased, bankruptcy, do-not-call...) member recommended; one member with two contacts the same day; a number in an explanation without a source; a surname, phone or email visible outside the bank; a released action that can still be changed; audit chain not verified; a collector's queue over their shift before any manager edits.

8 • Templates

8.1 Defect report

Defect ID	LM-____
Title	Short and specific: “Bulk approve includes escalated <i>HARDSHIP</i> rows”
Test case	e.g. DEC-06
Priority	P1 / P2 / P3 (section 2)
Environment	Build/date, Chrome version, Gemini key on/off, business date
Run id	run_...
Steps to reproduce	Numbered, starting from a known state (e.g. “after reset 3.4”)
Expected	Copy from the test case

Actual	What happened, exact message text
Reproducible	Always / Sometimes (x of y) / Once
Evidence	Screenshots, browser console lines, server traceback
Status	New → Assigned → Fixed → Retest → Closed / Reopened

8.2 Daily status (send at end of each test day)

Date / build:
Cases executed today: Pass: Fail: Blocked:
Cumulative: executed __ of 108 P1 passed __ of 55
New defects: P1 __ P2 __ P3 __ (list ids + titles)
Retested: fixed __ reopened __
Blockers:
Plan for tomorrow:

9 • Exit criteria and sign-off

☐	All 55 P1 cases executed and passed (retest pass counts)
☐	At least 90% of all 108 cases executed
☐	No open P1 defects; open P2 defects agreed with the product owner
☐	One full business day completed end to end: team → run → re-plan → decide → release → close → final report → next day carried forward
☐	Configuration files restored to their original values (CFG-09)
☐	Tracking sheet, evidence folder and defect list handed to the developer lead

Tester — name, signature, date

Developer lead — name, signature, date

Product owner — name, signature, date